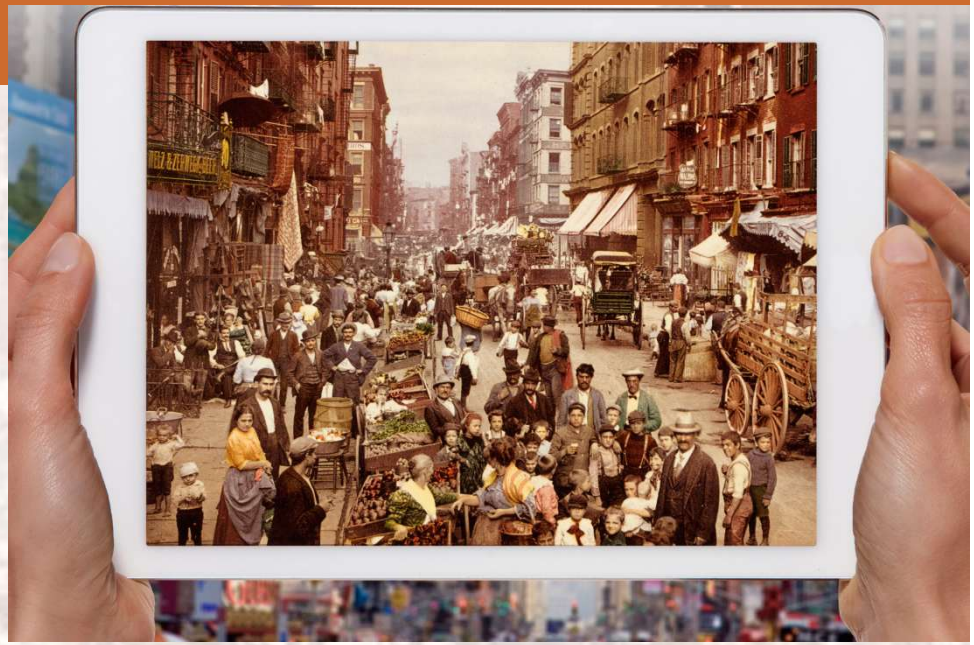
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CHAPTER 9

Application: International Trade

National Central University

Look for the answers to these questions:

- What determines how much of a good a country will import or export?
- Who benefits from trade? Who does trade harm? Do the gains outweigh the losses?
- If policymakers restrict imports, who benefits? Who is harmed? Do the gains from restricting imports outweigh the losses?
- What are some common arguments for restricting trade? Do they have merit?



Review from Chapter 3

- A country has a comparative advantage in a good
  - If it produces the good at lower opportunity cost than other countries
- Countries can gain from trade
  - If each exports the goods in which it has a comparative advantage



The Determinants of Trade

- The equilibrium without trade
  - Only domestic buyers and sellers
  - Equilibrium price and quantity
    - Determined on the domestic market
  - Total benefits
    - Consumer surplus
    - Producer surplus



World Price and Comparative Advantage

- P_W = the world price of a good, the price that prevails in world markets
- P_D = domestic price without trade

- If $P_D < P_W$,
  - Domestic country has comparative advantage, country exports the good
- If $P_D > P_W$,
  - Domestic country does not have comparative advantage, country imports the good



The Small Economy Assumption

- A small economy is a price taker in world markets:
  - Its actions have no effect on P_W
  - When a small economy engages in free trade, P_W is the only relevant price:
    - No seller would accept less than P_W (can sell the good for P_W in world markets)
    - No buyer would pay more than P_W (can buy the good for P_W in world markets)

A Country That Exports Soybeans

Without trade,

P_D = \$4

Q = 500

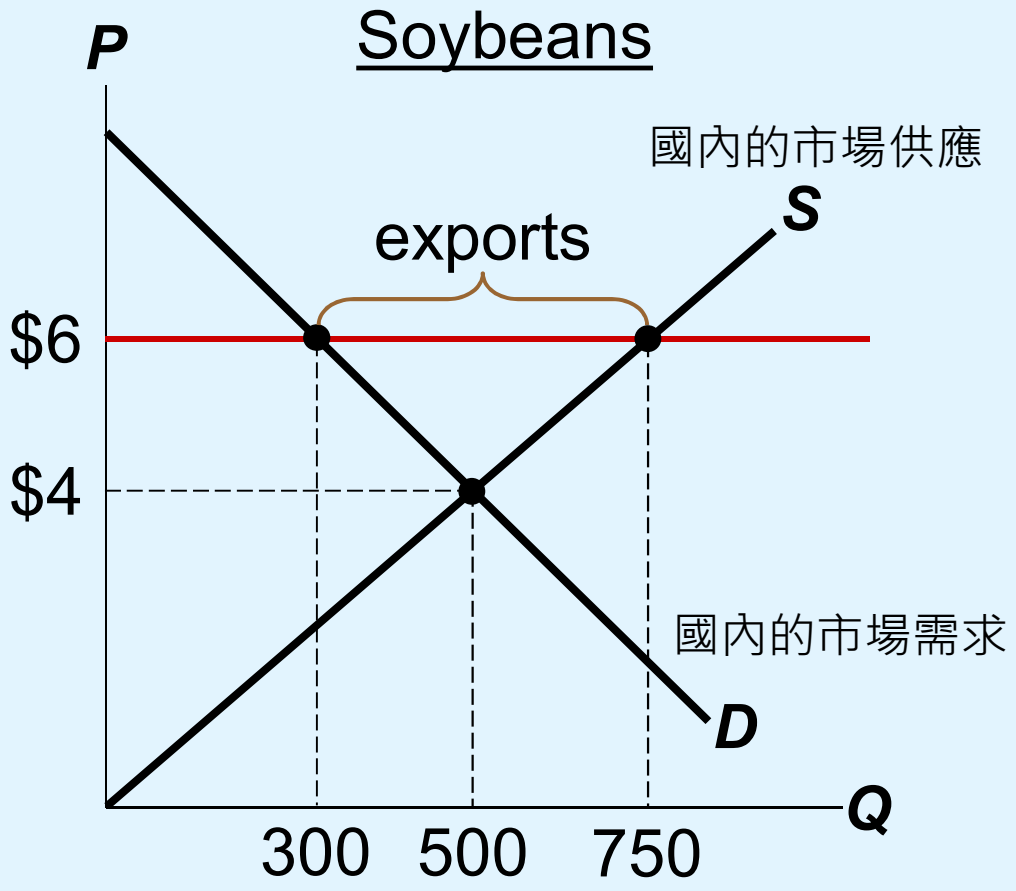
P_W = \$6

Under free trade,

domestic consumers demand 300

domestic producers supply 750

exports = 450



A Country That Exports Soybeans

Without trade,

CS = A + B

PS = C

Total surplus

= A + B + C

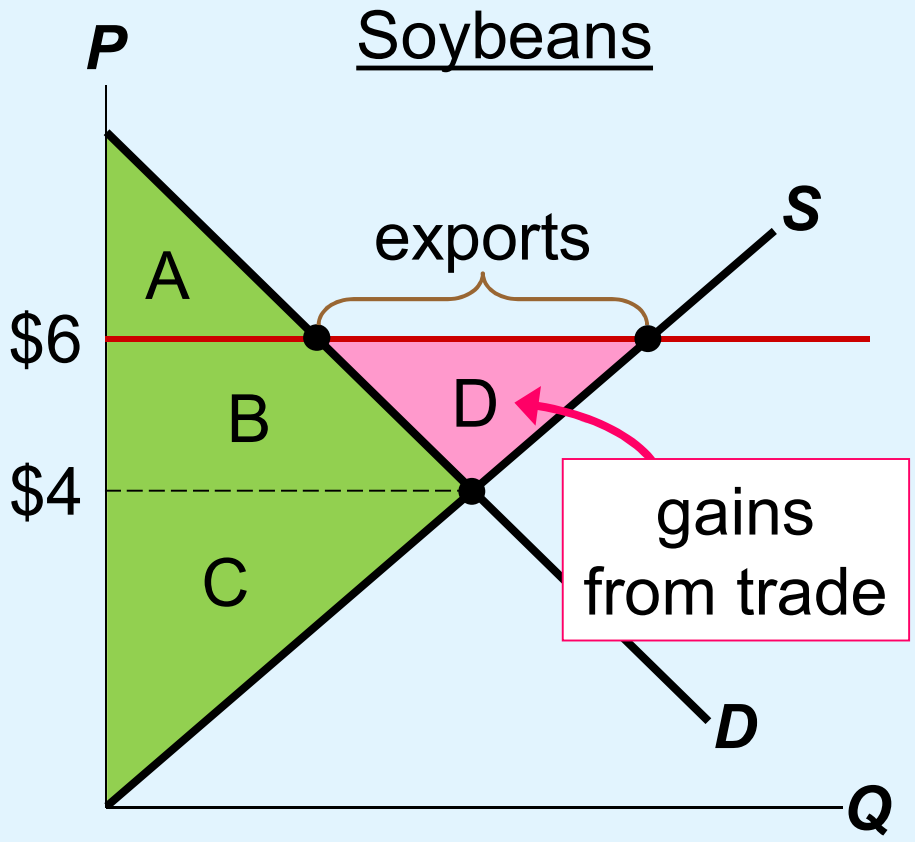
With trade,

CS = A

PS = B + C + D

Total surplus

= A + B + C + D



Active Learning 1

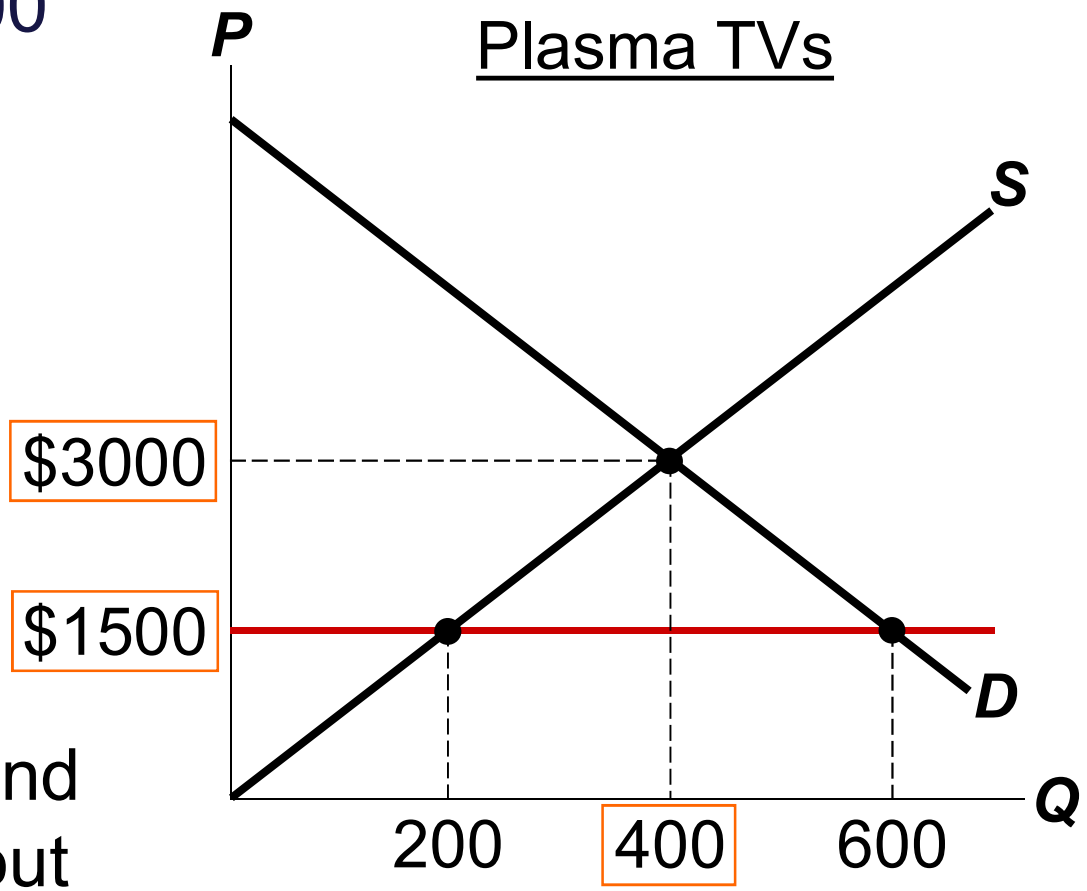
Analysis of Trade

Without trade,
P_D = \$3000, Q = 400

In world markets,
P_W = \$1500

- Under free trade, how many TVs will the country import or export?

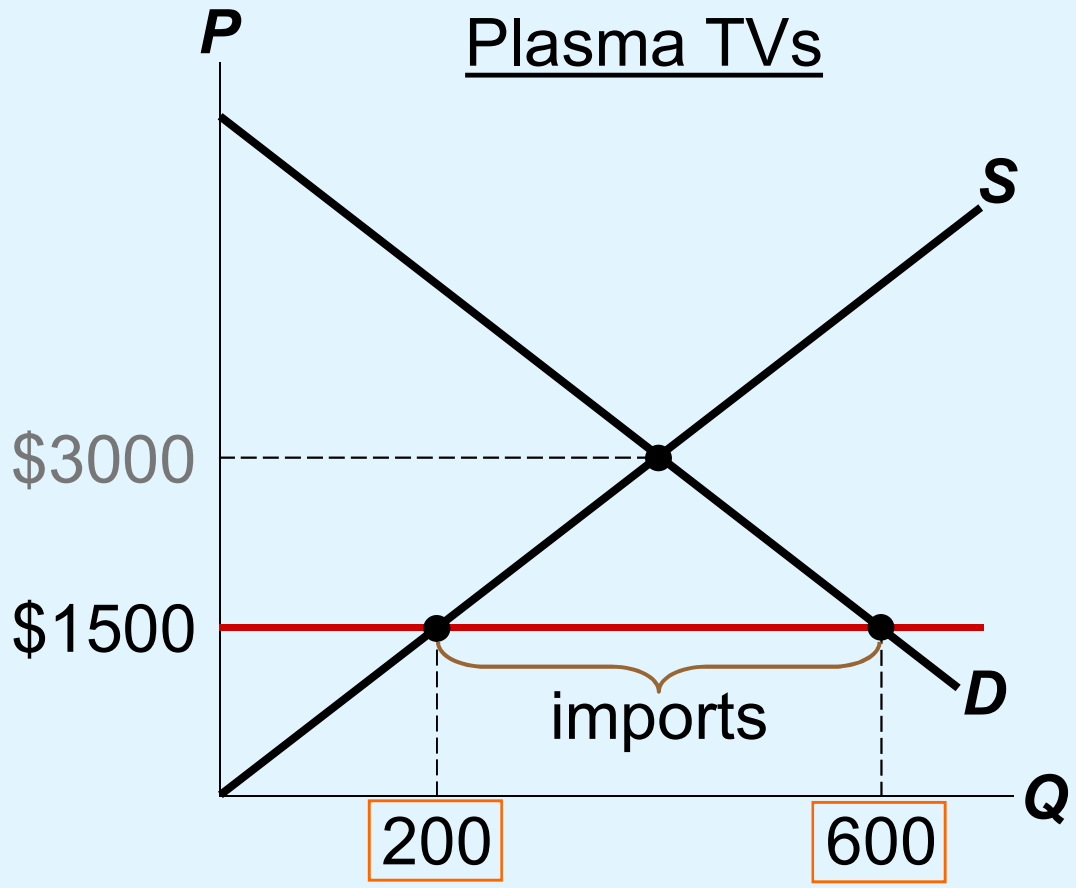
- Identify CS, PS, and total surplus without trade, and with trade.



Active Learning 1 Answers

Under free trade,

- domestic consumers demand 600
- domestic producers supply 200
- imports = 400



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Active Learning 1 Answers

Without trade,

CS = A

PS = B + C

Total surplus

= A + B + C

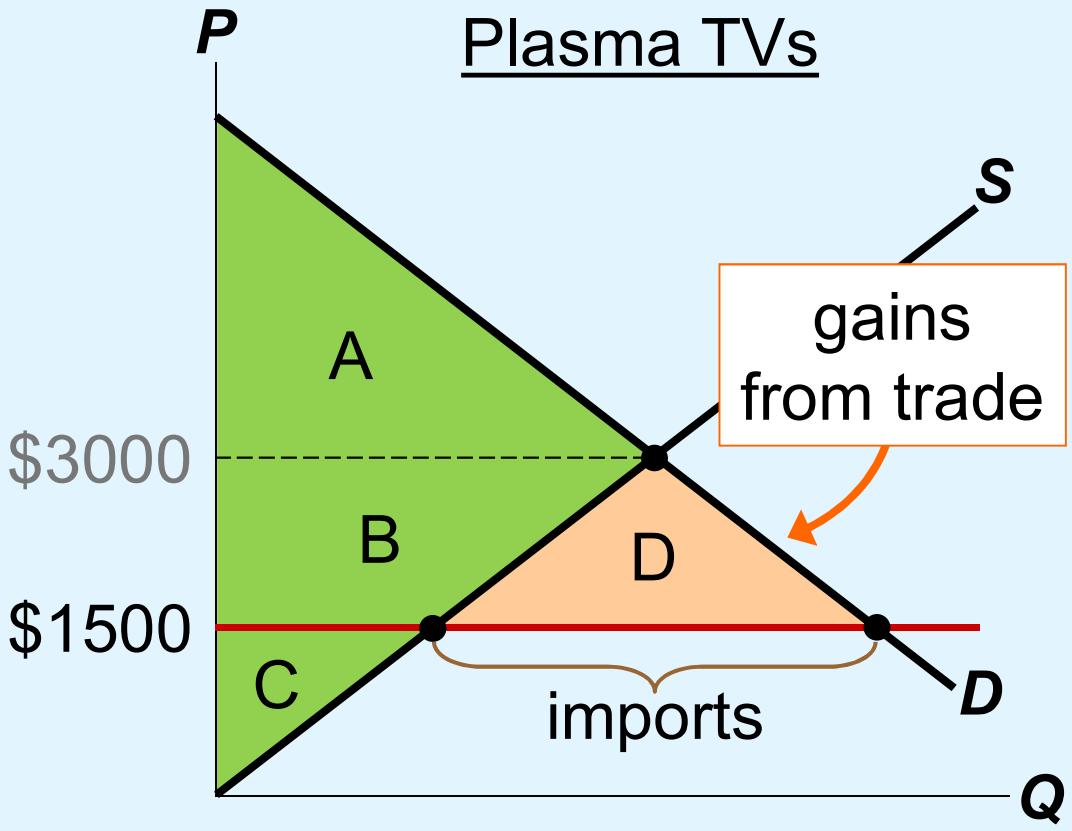
<u>With trade</u>,

CS = A + B + D

PS = C

Total surplus

= A + B + C + D



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Summary: The Welfare Effects of Trade

	$P_D < P_W$	$P_D > P_W$
direction of trade	exports	imports
consumer surplus	falls	rises
producer surplus	rises	falls
total surplus	rises	rises

Whether a good is imported or exported, trade creates winners and losers.

But the gains exceed the losses.

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Question

Consider a country that imports a good from abroad. For each of following statements, state whether it is true or false.

a. The greater the elasticity of demand, the greater the gains from trade

b. If demand is perfectly inelastic, consumers do not benefit form trade

Question

The world price of wine is below the price that would prevail in Canada in the absence of trade.

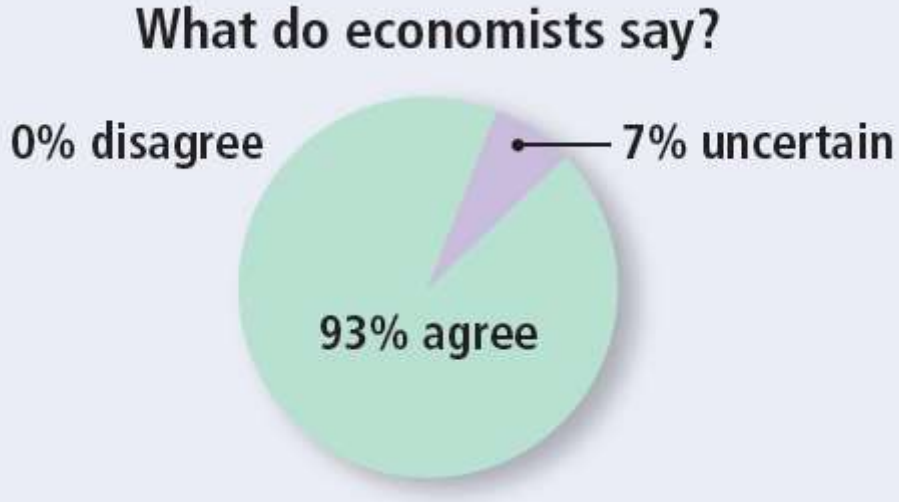
a. Assuming that Canadian imports of wine are a small share of total world wine production, draw a graph for the Canadian market for wine under free trade.

b. Now suppose that an unusual shift of the Gulf Stream leads to an unseasonably cold summer in Europe, destroying much of the grape harvest there.
   1. What effect does this shock have on the world price of wine?
   2. Show the effect on consumer surplus , producer surplus, and total surplus in Canada.
   3. Who are the winners and losers? Is Canada as a whole better or worse off?

ASK THE EXPERTS

Trade Deals

"Past major trade deals have benefited most Americans."





Winners and Losers From Trade

- Other benefits of international trade
  - Consumers: increased variety of goods
  - Producers: lower costs - economies of scale
  - Increased competition: reduce market power of domestic firms (increase total welfare)
  - Enhanced flow of ideas, facilitates the spread of technological advances around the world



Winners and Losers From Trade

- Then why all the opposition to trade?
  - The losers have more incentive to organize and lobby for restrictions on trade:
    - Losses: concentrated among a small group of people, who feel them acutely
    - Gains: spread thinly over many people, who may not see how trade benefits them
    - The winners from trade could compensate the losers and still be better off (such compensation rarely occurs)

1000萬人收益增加10元 vs. 100人收益減少10萬元



Winners and Losers From Trade

- Tariff
  - Tax on goods produced abroad and sold domestically
- Free trade
  - Domestic price = World price
- Tariff on imports
  - Raises domestic price above world price
    - By the amount of the tariff

國內買家與賣家面臨的價格為P_w+T

Analysis of a Tariff on Cotton Shirts

P_W = \$20

Free trade:

buyers demand 80

sellers supply 25

imports = 55

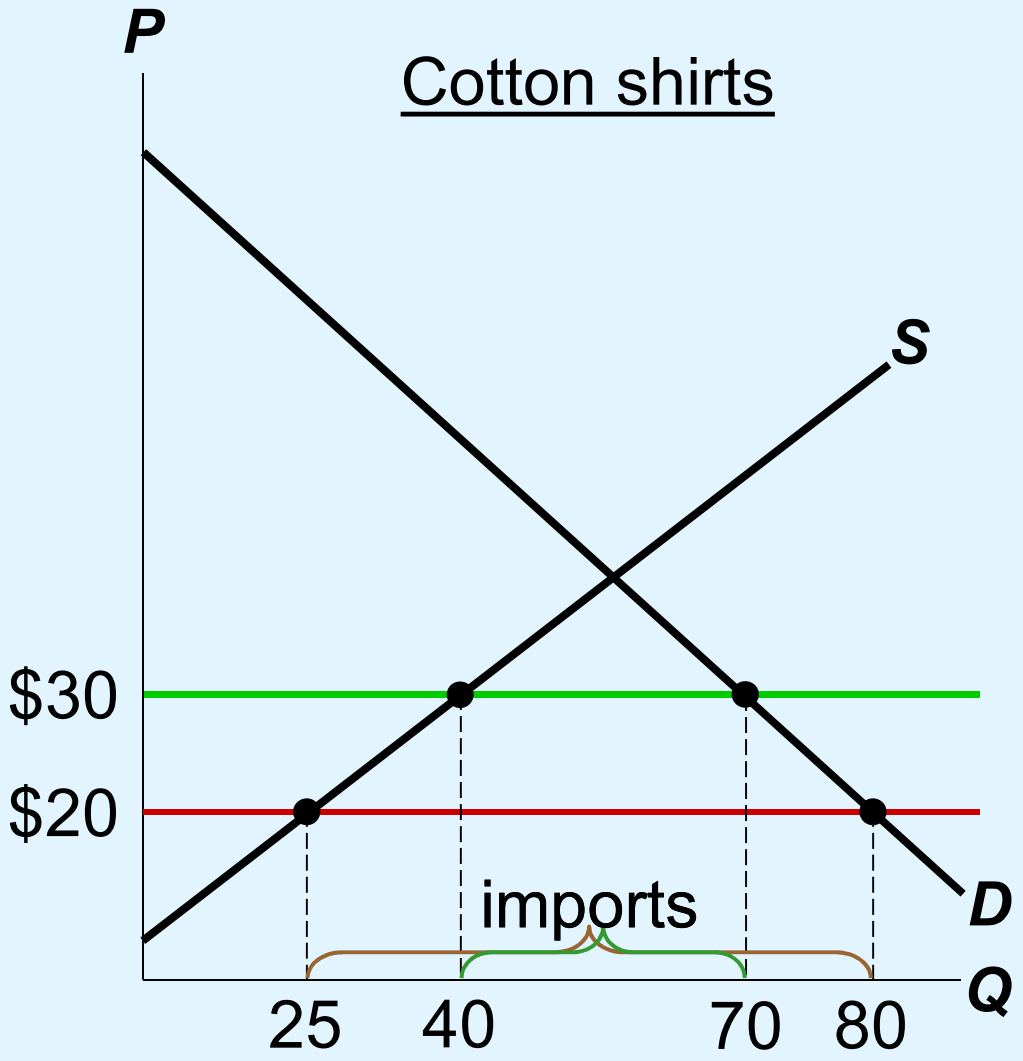
T = \$10/shirt

price rises to \$30

buyers demand 70

sellers supply 40

imports = 30



Analysis of a Tariff on Cotton Shirts

Free trade

CS = A + B + C
+ D + E + F

PS = G

Total surplus = A + B
+ C + D + E + F + G

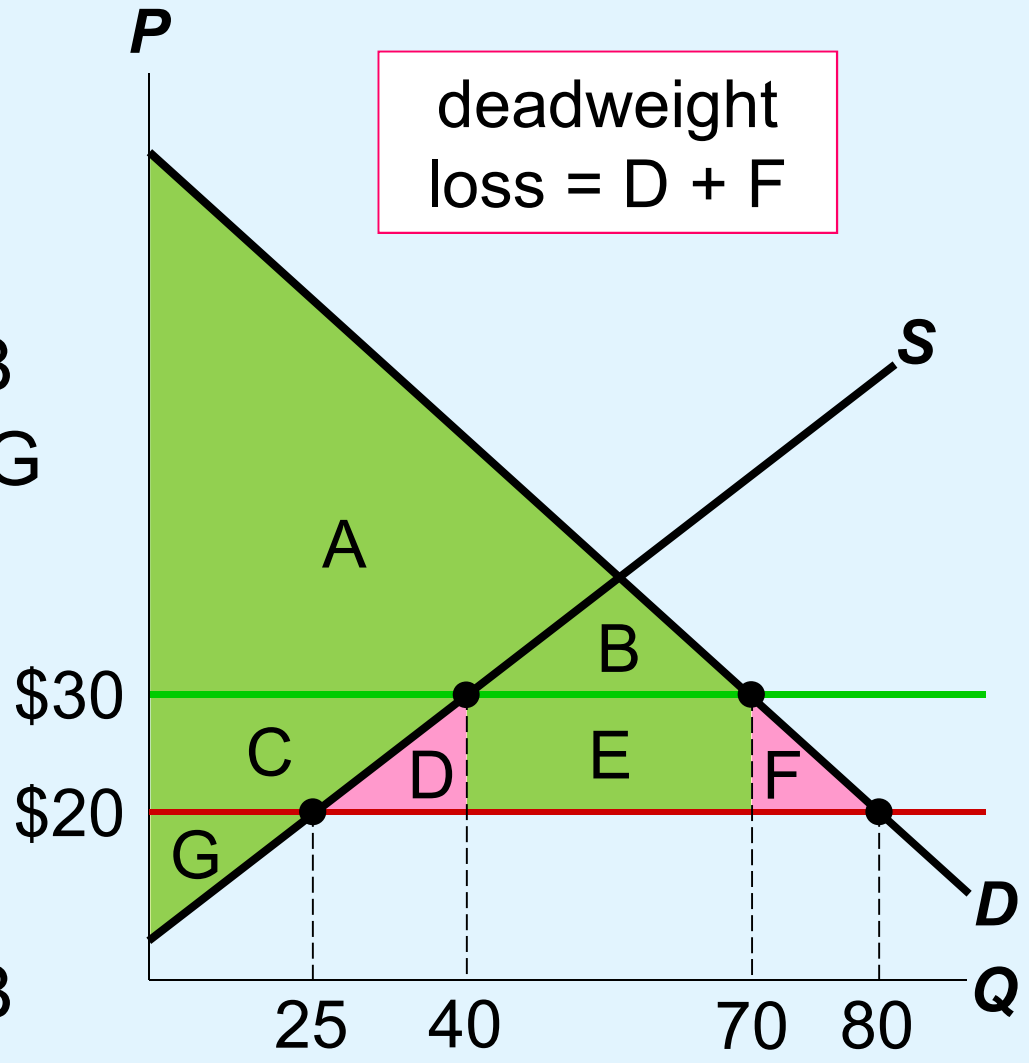
With tariff

CS = A + B

PS = C + G

Revenue = E

Total surplus = A + B
+ C + E + G



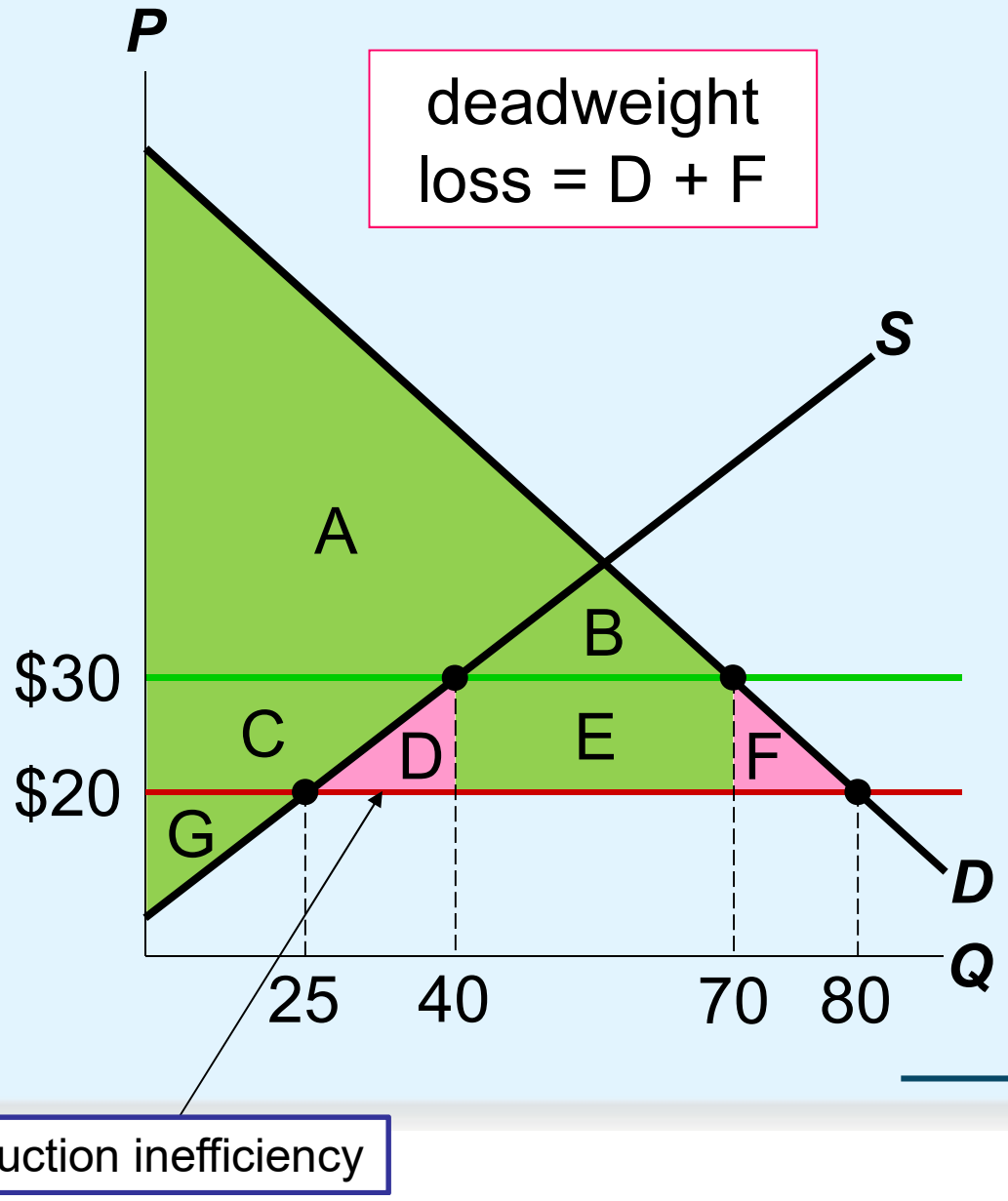
Analysis of a Tariff on Cotton Shirts

D = deadweight loss from the overproduction of shirts 過量生產

原本供應25單位增加到40單位

F = deadweight loss from the under-consumption of shirts 消費不足

原本消費80單位減少到70單位



Question

When China's clothing industry expands, the increase in world supply lowers the world price of clothing.

a. Draw an appropriate diagram to analyze how this change in price affects consumer surplus, producer surplus, and total surplus in a nation that imports clothing

b. Draw an appropriate diagram to analyze how this change in price affects consumer surplus, producer surplus, and total surplus in a nation that exports clothing

c. Compare your answers to parts (a) and (b).

Question

Having rejected a tariff on textiles (a tax on imports), the president of Isoland is now considering the same-sized tax on textile consumption (including both imported and domestically produced textiles).

a. Using Figure 4, identify the quantity consumed and the quantity produced in Isoland under a textile consumption tax.
b. Consider a table similar to that in Figure 4 for the textile consumption tax.
c. Which raises more revenue for the government—the consumption tax or the tariff? Which has a smaller deadweight loss? Explain.

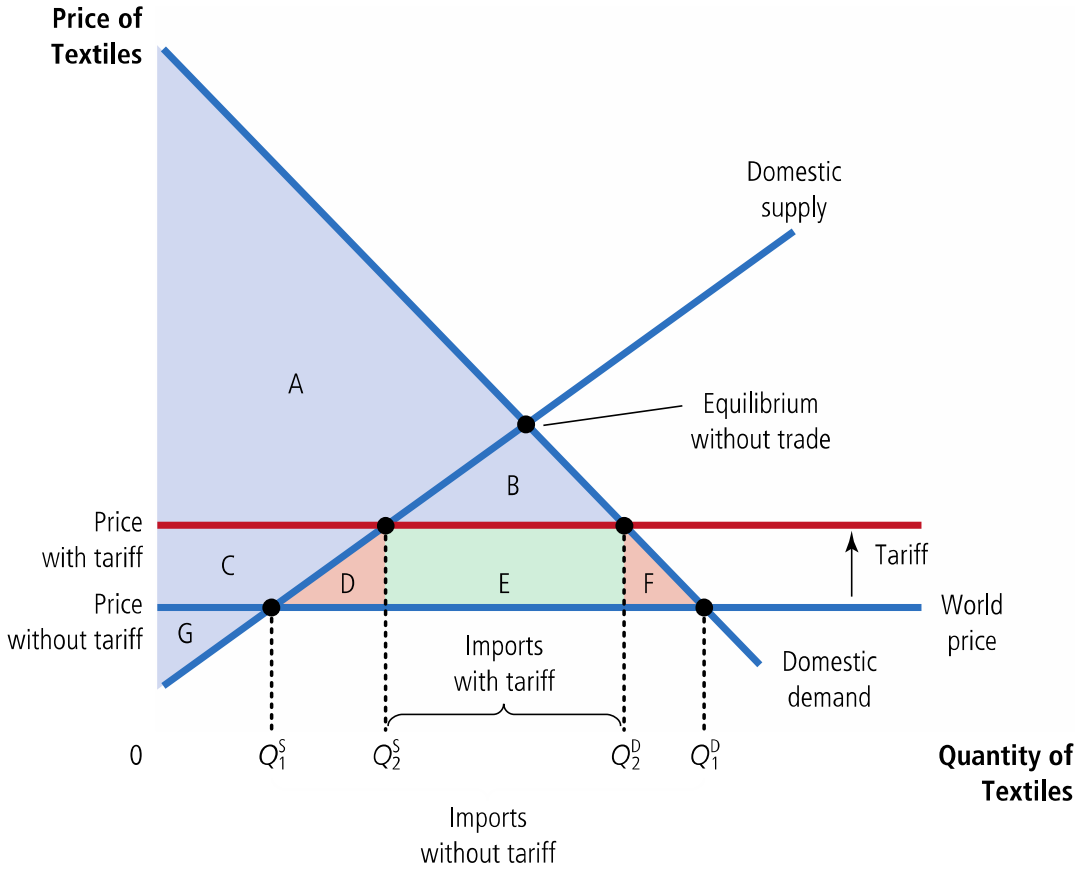
FIGURE 4

The Effects of a Tariff

A tariff, a tax on imports, reduces the quantity of imports and moves a market closer to the equilibrium that would exist without trade. Total surplus falls by an amount equal to area D + F. These two triangles represent the deadweight loss from the tariff.

	Before Tariff	After Tariff	Change
Consumer Surplus	A + B + C + D + E + F	A + B	−(C + D + E + F)
Producer Surplus	G	C + G	+ C
Government Revenue	None	E	+ E
Total Surplus	A + B + C + D + E + F + G	A + B + C + E + G	−(D + F)

The area D + F shows the fall in total surplus and represents the deadweight loss of the tariff.



Question ***

Consider a small country that exports steel. Suppose that a “pro-trade” government decides to subsidize the export of steel by paying a certain amount for each ton sold abroad.

- How does this export subsidy affect **the domestic price of steel**, **the quantity of steel produced**, **the quantity of steel consumed**, and **the quantity of steel exported**?
- How does it affect consumer surplus, producer surplus, government revenue, and total surplus?
- Is it a good policy from the standpoint of economic efficiency ?



Import Quotas

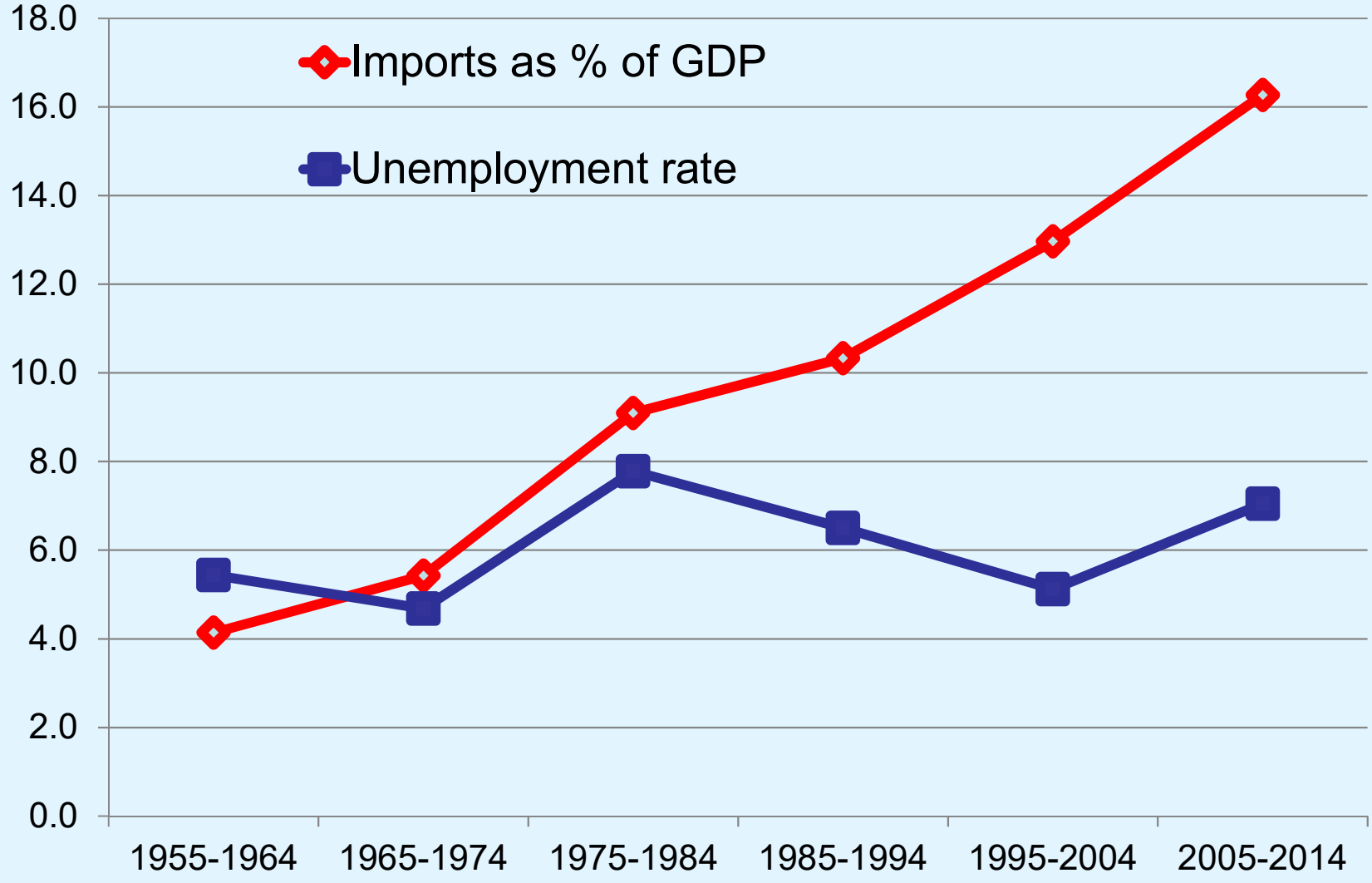
- Import quota
  - Quantitative limit on imports of a good
  - Mostly has the same effects as a tariff:
    - Raises price, reduces quantity of imports
    - Reduces buyers' welfare
    - Increases sellers' welfare
  - Creates profits for **the foreign producers of the imported goods**, who can sell them at higher price



Arguments For Restricting Trade

- The jobs argument
  - "Trade with other countries destroys domestic jobs"
  - Free trade creates jobs at the same time that it destroys them
  - Total unemployment does not rise as imports rise, because job losses from imports are offset by job gains in export industries….

U.S. Imports & Unemployment, decade averages, 1955–2014





Arguments For Restricting Trade

- The national-security argument
  - "The industry is vital for national security and it should be protected from foreign competition, to prevent dependence on imports that could be disrupted during wartime"
  - When there are legitimate concerns over national security
    - But producers may exaggerate their own importance to national security to obtain protection from foreign competition



Arguments For Restricting Trade

- The infant-industry argument
  - "New industries need temporary trade restriction to help them get started"
  - Difficult to implement in practice
  - The temporary policy is hard to remove
  - Protection is not necessary for an infant industry to grow



Arguments For Restricting Trade

- The unfair-competition argument
  - "Producers argue their competitors in another country have an unfair advantage, e.g. due to government subsidies"
  - Increase in total surplus for the country
    - We should welcome imports of low-cost products subsidized by the other country's taxpayers
    - The gains to our consumers will exceed the losses to our producers



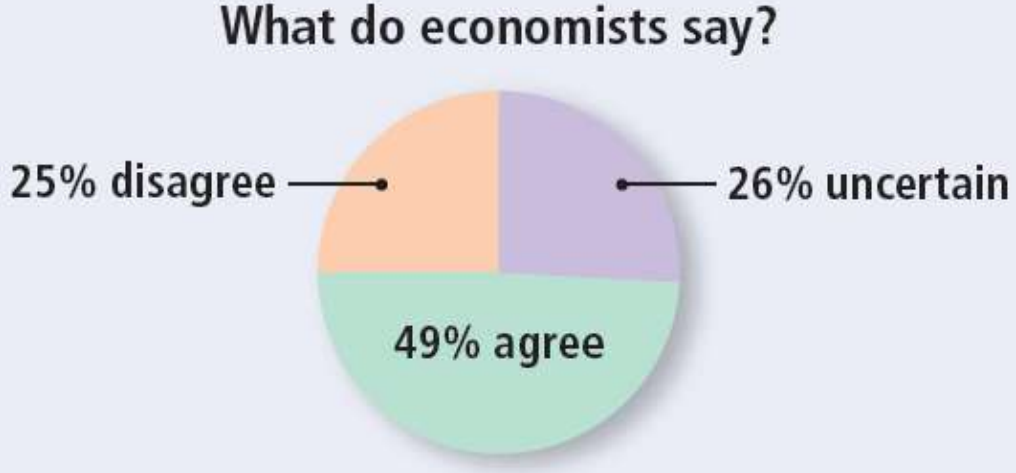
Arguments For Restricting Trade

- The protection-as-a-bargaining-chip argument
  - "Trade restrictions can be useful when we bargain with our trading partners"
  - The threat may not work

ASK THE EXPERTS

Trade Deals

"Refusing to liberalize trade unless partner countries adopt new labor or environmental rules is a bad policy, because even if the new standards would reduce distortions on some dimensions, such a policy involves threatening to maintain large distortions in the form of restricted trade."



Trade agreements and the WTO

- World Trade Organization, WTO
- Unilateral approach to achieve free trade
  - Remove its trade restrictions on its own
  - Great Britain, 19th century
  - Chile and South Korea, recent years
- Multilateral approach to free trade
  - Reduce its trade restrictions while other countries do the same
  - NAFTA(北美自由貿易協定), GATT(關稅及貿易總協定)

Trade agreements and the WTO

- North American Free Trade Agreement (NAFTA;北美自由貿易協定)
  - 1993, lowered trade barriers among the United States, Mexico, and Canada
- General Agreement on Tariffs and Trade (GATT;關稅及貿易總協定)
  - Continuing series of negotiations among many of the world's countries with the goal of promoting free trade

GATT其實是個暫時協定，於1948年1月1日生效，直到1993年1月1日它的條文成為世界貿易組織(WTO)架構下之一部分為止

CASE STUDY

Trade agreements and the WTO

- GATT
  - **United States helped to found GATT**
    - After World War II
    - In response to the high tariffs imposed during the Great Depression
  - Successfully reduced the average tariff among member countries from about 40% to 5%
  - Enforced by the WTO
  - 2015: 162 countries; more than 97 % of world trade



Trade agreements and the WTO



- Advantages of the multilateral approach
  - Potential to result in freer trade than unilateral approach
    - Reduce trade restrictions abroad and at home
  - Political advantage
    - Producers are fewer and better organized than consumers
    - Greater political influence

Summary

- A country will export a good if the world price of the good is higher than the domestic price without trade. Trade raises producer surplus, reduces consumer surplus, and raises total surplus.
- A country will import a good if the world price is lower than the domestic price without trade. Trade lowers producer surplus but raises consumer and total surplus.
- A tariff benefits producers and generates revenue for the government, but the losses to consumers exceed these gains.

Summary

- Common arguments for restricting trade include: protecting jobs, defending national security, helping infant industries, preventing unfair competition, and responding to foreign trade restrictions.
- Some of these arguments have merit in some cases, but economists believe free trade is usually the better policy.

Who Wins and Who Loses from Globalization?



Eye on Globalization (1 of 4)

Who Wins and Who Loses from Globalization?

Economists generally agree that the gains from globalization vastly outweigh the losses.

But there are both winners and losers.

The U.S. consumer is a big winner.

Globalization has brought iPads, Wii games, Nike shoes, and a wide range of other products to our shops at ever lower prices.



Eye on Globalization (2 of 4)

Who Wins and Who Loses from Globalization?

The Indian (and Chinese and other Asian) worker is another big winner.

Globalization has brought a wider range of more interesting jobs and higher wages.



Eye on Globalization (3 of 4)

Who Wins and Who Loses from Globalization?

The U.S. (and European) **textile workers** and **furniture makers** are big losers.

Their jobs have disappeared and many of them have struggled to find new jobs even when they've been willing to take a pay cut.



Eye on Globalization (4 of 4)

Who Wins and Who Loses from Globalization?

But one of the *biggest* losers is the **African farmer**.

Blocked from global food markets by trade restrictions and subsidies in the United States and Europe, globalization is leaving much of Africa on the sidelines.

